

RANE GRAY

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EDUCATION

University of Colorado Boulder | *B.S. in Computer Science* | Expected Spring 2027 | GPA: 3.82

TECHNICAL SKILLS

Robotics: ROS 2, digital twins, simulation (NVIDIA Isaac Sim / Omniverse), OpenUSD, URDF, inverse kinematics, SLAM / 2D mapping, sensor integration (LiDAR / RGB-D), teleoperation

Programming: Python, C++, JavaScript / TypeScript, Bash, Git, Linux/Unix, SQL, React, Node.js

ML / Perception: synthetic data generation, imitation learning (SmolVLA), computer vision (OpenCV / Open3D)

Software: Testing, CI/CD, GitLab, Jira, technical documentation

EXPERIENCE

Robotics Engineering Intern, Ricoh USA | May–Aug 2026 · continuing part-time

Digital Twin, Robotics Team

- Built a physics-based digital twin of an industrial palletizing cell in Isaac Sim / Omniverse, converting production STEP assets into a tuned USD scene driven by the real control software unmodified, removing the physical cell as the only place changes could be tested.
- Validated *placement* accuracy by hand across 101 parts, 4 runs, and both pallet positions: ~2 mm mean error (max < 7 mm) against a 10 mm job tolerance, repeatable to ~0.1 mm, establishing the twin as a trusted test target.
- Established regression CI for the twin, diffing automated Isaac Sim runs against known-good baselines on validation-derived thresholds so every change is testable before it reaches hardware; added LLM-based code review on merge requests.
- Building a synthetic data generation pipeline from simulation to produce labeled training data for vision models.

Undergraduate Research Assistant, Correll Lab | Feb 2025–Present

University of Colorado Boulder

- Developed a head/neck pan-tilt subsystem for the Unitree H1-2 humanoid, from mechanism design through ROS 2 control and platform validation.
- Implemented an Apple Vision Pro teleoperation interface for the H1-2 with head and pose tracking, building on the *Open-Television* line of work.

F-16 Avionics Systems Expert, USAF & COANG | Aug 2015–Aug 2025

Arizona, South Korea, Germany, Colorado

- Led a 23-person avionics team diagnosing, repairing, and validating mission-critical electronic systems, sustaining 95% mission-ready aircraft under tight schedules.
- Designed a standardized troubleshooting and training pipeline (checklists, sign-offs, proficiency gates) that cut qualification time 30% while improving consistency and safety.

PUBLICATIONS & AWARDS

Cutting the Cord: System Architecture for Low-Cost, GPU-Accelerated Bimanual Mobile Manipulation | arXiv:2603.09051, 2026

Shaw, Liu, Costa, **Gray**, Skowronek, Diaz, Bui, Correll

Hugging Face x LeRobot Hackathon, 1st Place Team | June 2025

SmolVLA fine-tuned on 50 teleoperated demonstrations (SO-101) and evaluated on hardware.

PROJECTS

Senior Thesis: Agentic Test-Time Compute for Low-Cost Robotics | Aug 2026–May 2027

Advisor: Prof. Nikolaus Correll

- Investigating whether agentic retry, visual differencing, and structured feedback can recover the task reliability lost to low-cost actuators, and at what token cost relative to better hardware.
- Building an instrumented harness on xLeRobot that logs per-attempt outcomes and token spend across hardware configurations and retry budgets.

Click & Whirr, clickandwhirr.com | 2025–Present

- Built and shipped a browser-based platform where learners write real Python to drive a simulated robot, spanning a 13-lesson curriculum from line following and PID control through odometry, path planning, and state estimation.